

WGNI FARM

WATER MANAGEMENT STANDARD OPERATING PROCEDURE

For Greenhouse and Other Site Usage

Field	Detail
SOP Reference	WGF-OPS-WTR-001
Subject	Water Management Greenhouse, Block Moulding & All Site Usage
Location	WGNI Farm, Abeokuta
Prepared By	Farm Operations Management
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Section A System Overview

A1. Purpose

This Standard Operating Procedure governs the management, distribution, and usage of water across all active operational zones at WGNU Farm, Abeokuta. It covers the full water chain from borehole source through to end use across greenhouse irrigation, block moulding construction, poultry, fishpond, and any other site activity drawing from the farm water supply. The goal is to ensure water is used efficiently, tracked consistently, and that the greenhouse irrigation supply is always protected as the highest priority demand.

A2. Water Chain Sequence

Water at WGNU Farm flows through a defined sequence from source to end use. Every stage must be functional before the next can operate effectively.

BOREHOLE → INTAKE PUMPS (Ecoflow Solar / Electric Surface Pump) → 6 x 5,000L BASE TANKS (30,000L) → DUAL FILTRATION UNIT → MAIN WATER TOWER (4 x 2,500L Geepee = 10,000L) → EN-ROUTE CHANNELS → STAND 1 & STAND 2 STOREX TANKS (4,000L) → DRIP IRRIGATION LINES → CROP STANDS

Block moulding, poultry, fishpond, and construction draw directly from the base tanks via hose pipe they do not pass through the filtration unit.

A3. Full Infrastructure Inventory

Component	Specification	Quantity
Borehole	Primary water source	1
Ecoflow hybrid solar pump	Primary intake solar + AC generator backup	1
Electric surface pump	Secondary/backup intake AC generator powered	1
Generator (gen-set)	Powers electric surface pump + Ecoflow AC backup	1
Base storage tanks	5,000L each = 30,000L total	6 units
Dual filtration vessels	Sand filter + carbon filter	1 set (2 vessels)
Inline pre-filter cartridges	Blue cylindrical, at filter base	2 units
Electric booster pump	Drives water through filter vessels	1
2HP petrol transfer pump	Transfers filtered water to main water tower	1
HDPE pipe	Blue-stripe, borehole to base tanks	As installed
Green HDPE distribution pipe	Filter unit to main tower and greenhouse	As installed
Main water tower	Tall red steel frame structure	1
Main overhead tanks (Geepee)	4 x 2,500L = 10,000L total	4 units
Stand 1 (Storex)	2 x 1,000L = 2,000L feeds Kit 1, 2, 3, 6	1 stand
Stand 2 (Storex)	2 x 1,000L = 2,000L feeds Kit 4, 5, 7	1 stand

Total overhead capacity	Main tower + Stand 1 + Stand 2	14,000L
Hose pipe block moulding	Direct draw from base tanks to moulding site	1 run
Block moulding machine	Motorised petrol engine	1
Active greenhouse kits	Kit 1 (500 sqm, 1,200 stands) Kit 7 (250 sqm, 600 stands)	2 kits

A4. Greenhouse Kit Assignment

Structure	Capacity	Kits Served	Kit Size	Active
Main water tower	10,000L	All kits		Yes
Stand 1 (Storex)	2,000L	Kit 1, 2, 3, 6	Kit 1–5: 500sqm / Kit 6: 250sqm	Kit 1
Stand 2 (Storex)	2,000L	Kit 4, 5, 7	Kit 4–5: 500sqm / Kit 7: 250sqm	Kit 7
Total overhead	14,000L			

A5. Pumping Modes

Mode	Route	Purpose
Mode 1	Transfer pump → Filter → Main tower → gravity cascade to Stand 1 & Stand 2	Standard full system fill
Mode 2	Transfer pump → Filter → Stand 1 only (en-route channel)	Direct fill Stand 1
Mode 3	Transfer pump → Filter → Stand 2 only (en-route channel)	Direct fill Stand 2

A6. Other Site Water Usage

Site	Source	Method	Priority
Greenhouse irrigation	Main tower → Storex stands → drip tape	Gravity	Highest
Block moulding	Base tanks direct	Hose pipe	Secondary
Poultry	Base tanks direct	As required	Secondary
Fishpond	Base tanks direct	As required	Secondary
Construction (general)	Base tanks direct	As required	Secondary

Section B Water Source & Base Tank Filling

B1. Primary Intake Ecoflow Hybrid Solar Pump

The Ecoflow hybrid solar pump is the primary and preferred method for filling the base tanks. It draws directly from the borehole and pumps into the 6 x 5,000L base tanks. It operates on two power sources managed automatically solar panels during daylight hours and AC generator power as backup when solar irradiance is insufficient. One generator serves both the Ecoflow AC backup input and the electric surface pump.

Operating Procedure

1. Confirm solar panels are clean and unobstructed every morning.
2. Check the Ecoflow controller on the overhead control panel confirm it shows active charging and pump status.
3. The system operates automatically during daylight hours typically 8:00 AM to 5:30 PM switching between solar and generator AC backup as needed.
4. Monitor base tank levels morning and evening to confirm the system is maintaining adequate storage.
5. If the controller shows a fault or tanks are not rising despite good sunlight and generator running, stop the system and escalate to the Farm Manager immediately.

Power Priority

- Solar input: primary zero fuel cost
- Generator AC input: backup activates automatically when solar is insufficient

B2. Secondary Intake Electric Surface Pump

The electric surface pump is a standalone centrifugal pump powered directly by the generator via AC cable. It is used when the Ecoflow system alone cannot maintain adequate base tank levels during extended cloudy periods, high-demand days, or when the Ecoflow unit is under maintenance.

Operating Procedure

1. Confirm the generator is running and stable before starting the electric surface pump.
2. Check that the electrical cable connection from generator to pump is secure.
3. Confirm the inlet pipe is fully submerged in the borehole.
4. Confirm the outlet pipe is directed into the base tank inlet.
5. Switch on the electric surface pump at the control switch.
6. Confirm water flowing into the base tank within 90 seconds.
7. Monitor continuously do not leave unattended.
8. Switch off pump before stopping the generator.

B3. Generator Operating Rules

The generator serves two simultaneous roles it supplies AC backup power to the Ecoflow unit and directly powers the electric surface pump.

- Always start the generator before switching on the electric surface pump.
- Never stop the generator while the electric surface pump is still running.
- Check generator fuel level at the start of every operating day.
- Allow the generator to warm up for 2 minutes before connecting load.
- Log generator run hours and fuel consumption in the daily operations record.
- Do not overload the generator by running both the electric surface pump and other heavy equipment simultaneously without confirming generator capacity.

B4. Base Tank Management Rules

Rule	Detail
Minimum safe level	Never allow combined base tank storage to drop below 5,000L
Daily check	Visually assess all 6 tanks every morning before any pumping or site activity begins
Solar-first	Allow Ecoflow solar system to run first. Use electric pump only when solar insufficient
Tank rotation	Monitor all 6 tanks do not draw repeatedly from one while others remain full
Contamination	Cover all base tank inlets when not in use. No surface runoff, debris, or animals near openings
Priority rule	If base tanks approach 5,000L pause ALL non-greenhouse draws immediately

Section C Filtration System

C1. System Description

The dual-vessel filtration unit sits between the base tanks and the main water tower. All water passing to the greenhouse must go through this system under standard operation. The unit is housed under the solar canopy structure adjacent to the base tanks.

- Two large blue FRP filter vessels (sand filter + activated carbon filter)
- Two inline pre-filter cartridge housings at the base of the vessels
- One electric booster pump that drives water pressure through the vessels
- A manifold valve system controlling service, backwash, and bypass modes

C2. Normal Service Operation

1. Confirm all manifold valves are in SERVICE position not backwash or bypass.
2. Switch on the electric booster pump.
3. Allow 2 minutes for pressure to build through the filter vessels before starting the transfer pump.
4. Confirm clean water is flowing at the filter outlet toward the main tower.
5. The filter unit must run continuously during any pumping session do not switch off mid-session.
6. Switch off the booster pump only after the transfer pump has been fully stopped.

C3. Filter Backwash Procedure

Backwash is required when flow rate drops noticeably or water clarity changes. Assess at least once per week during active irrigation periods.

Signs That Backwash is Due

- Noticeably reduced flow at the overhead tank despite pump running normally
- Discoloured or cloudy water at the filter outlet
- Unusual pressure drops across the filter vessels

Backwash Procedure

1. Stop the transfer pump first.
2. Switch manifold valves to BACKWASH position.
3. Run the booster pump for 10–15 minutes to flush accumulated sediment.
4. Switch to RINSE position for 3–5 minutes.
5. Return valves to SERVICE position.
6. Resume normal operation.
7. Log the backwash date and observations in the daily record.

C4. Pre-filter Cartridge Maintenance

Inspect inline blue cartridge pre-filters every two weeks. Replace when visibly clogged or discoloured. Always keep at least one spare set on the farm.

Section D Main Water Tower & Overhead Tank Filling

D1. Main Water Tower Description

The main water tower is a tall red steel frame structure holding 4 Geepee tanks arranged across two platform levels 2 tanks on the upper platform and 2 tanks on the lower platform. Total capacity is 10,000L. The tower is located approximately 80–100 metres from the pump and filtration station. Water pumped to the main tower gravity-feeds downward and outward through the distribution pipe network. En-route valve channels along the distribution line allow the operator to divert water into Stand 1 or Stand 2 Storex tanks before water reaches the full tower distance.

D2. Reference Parameters

Parameter	Value
Transfer pump	2HP petrol centrifugal pump
Pipe material	HDPE blue-stripe + green distribution pipe
Total head / pipe run to main tower	~80–100 metres
Effective flow rate	45–55 litres/minute
Fuel type	Petrol (Gasoline)
Fuel price	N1,300/litre
Fuel consumption rate	~0.55 litres/hour

D3. Pre-Operation Checks All Modes

- Confirm base tank level is above 5,000L.
- Check transfer pump engine oil at full mark, clean.
- Confirm petrol allocation for the intended session.
- Inspect HDPE pipe connections no cracks, loose unions, or leaks.
- Confirm filtration booster pump is operational.
- Confirm destination tank is ready to receive water.
- Confirm en-route channel valves are set correctly for the intended mode.
- Prime the pump if it has been idle for more than 24 hours.

D4. Mode 1 Fill Main Tower (Standard Full System Fill)

Volume	Est. Time	Petrol	Cost
10,000L	~185–220 min (~3 hrs)	~1.9L	~N2,470

1. Set all en-route channel valves to OPEN to allow cascade filling of Storex stands en route.
2. Start filtration booster pump. Allow 2 minutes to pressurise.
3. Start transfer pump using standard start-up procedure (see D7).
4. Confirm water flowing toward the main tower within 90 seconds.
5. Monitor every 20–30 minutes.
6. Stop pump when main tower tanks are full.
7. Switch off booster pump after transfer pump is stopped.
8. Log session in daily operations record.

D5. Mode 2 Fill Stand 1 Only (Kit 1, 2, 3, 6)

Volume	Est. Time	Petrol	Cost
2,000L	~35–40 min	~0.38L	~N494

1. Set en-route channel valve for Stand 1 to OPEN. Close valve beyond Stand 1.
2. Start filtration booster pump. Allow 2 minutes.
3. Start transfer pump.
4. Confirm water rising in Stand 1 Storex tanks within 90 seconds.
5. Monitor every 15 minutes.
6. Stop pump when Stand 1 tanks are full (~35–40 minutes).
7. Switch off booster pump. Log session.

D6. Mode 3 Fill Stand 2 Only (Kit 4, 5, 7)

Volume	Est. Time	Petrol	Cost
2,000L	~35–40 min	~0.38L	~N494

Same procedure as Mode 2. Confirm en-route channel valve for Stand 2 is OPEN and valve beyond Stand 2 is closed before starting.

D7. Standard Pump Start-Up Procedure

1. Set fuel petcock to ON.
2. Set choke to CLOSED (cold start) or OPEN (warm engine).
3. Set throttle to mid/start position.
4. Pull recoil starter firmly until engine fires.
5. Gradually open choke to fully OPEN once engine is running.
6. Allow 2–3 minutes warm-up before putting engine under full load.
7. Confirm water flow within 90 seconds of full operation.
8. If no flow after 2 minutes stops the engine and re-prime before attempting again.

D8. Standard Pump Shutdown Procedure

1. Reduce throttle gradually do not cut off abruptly.
2. Turn fuel petcock to OFF.
3. Allow engine to idle down and stop naturally.
4. Close inlet and outlet valves.
5. Allow 5 minutes cooling before covering or moving the pump.

Section E Greenhouse Irrigation & Water Consumption

E1. Active Kit Summary

Kit	Area	Plant Stands	Fed By	Overhead Capacity
Kit 1	500 sqm	1,200 stands	Main tower → Stand 1	2,000L (Storex)
Kit 7	250 sqm	600 stands	Main tower → Stand 2	2,000L (Storex)
Total active	750 sqm	1,800 stands		4,000L

E2. Inactive Kit Reference

Kit	Area	Fed By	Status
Kit 2	500 sqm	Stand 1	Inactive standby
Kit 3	500 sqm	Stand 1	Inactive standby
Kit 4	500 sqm	Stand 2	Inactive standby
Kit 5	500 sqm	Stand 2	Inactive standby
Kit 6	250 sqm	Stand 1	Inactive standby
Total inactive	2,000 sqm		Not yet planted

E3. Irrigation Volume Per Stand Per Day

Growth Stage	Litres Per Plant Stand Per Day
Transplant to 2 weeks (current)	0.5L
Early vegetative (2–5 weeks)	0.75L
Active vegetative / pre-flowering	1.0L
Flowering to early fruiting	1.25L
Full fruiting stage	1.5L

E4. Daily Water Consumption Per Kit

Current Stage Transplant to 2 Weeks

Kit	Stands	L/Stand/Day	Daily Consumption
Kit 1	1,200	0.5L	600L/day
Kit 7	600	0.5L	300L/day
Combined	1,800		900L/day

Early Vegetative Stage

Kit	Stands	L/Stand/Day	Daily Consumption
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Kit 1	1,200	0.75L	900L/day
Kit 7	600	0.75L	450L/day
Combined	1,800		1,350L/day

Full Fruiting Stage

Kit	Stands	L/Stand/Day	Daily Consumption
Kit 1	1,200	1.5L	1,800L/day
Kit 7	600	1.5L	900L/day
Combined	1,800		2,700L/day

E5. Valve Open Time Per Irrigation Cycle

Assumptions

Parameter	Value	Basis
Drip emitter flow rate	2 litres/hour per stand	Standard agricultural drip tape Nigerian greenhouse
Irrigation cycles per day	2 cycles	Confirmed 6:30 AM and 4:30 PM
Valve configuration	Separate Kit 1 and Kit 7 run independently	Confirmed

Calculation Method

Valve open time (minutes) = Volume needed per cycle divided by (Emitter flow rate per stand x Number of stands) x 60

Water Per Cycle Current Stage (Transplant)

Kit	Daily Need	Per Cycle (divided by 2)	System Flow Rate	Valve Open Time
Kit 1	600L/day	300L per cycle	2L/hr x 1,200 = 2,400L/hr	7.5 minutes
Kit 7	300L/day	150L per cycle	2L/hr x 600 = 1,200L/hr	7.5 minutes

Valve Open Time Schedule All Growth Stages

Growth Stage	L/Stand/Day	Kit 1 Minutes per cycle	Kit 7 Minutes per cycle
Transplant (current)	0.5L	7.5 min	7.5 min
Early vegetative	0.75L	11.25 min	11.25 min
Active vegetative	1.0L	15 min	15 min
Flowering / early fruiting	1.25L	18.75 min	18.75 min
Full fruiting	1.5L	22.5 min	22.5 min

Daily Valve Operation Schedule Current Stage

Kit	Cycle 1 (6:30 AM)	Cycle 2 (4:30 PM)	Total Valve Open Time/Day
Kit 1	Open 7.5 minutes	Open 7.5 minutes	15 min/day
Kit 7	Open 7.5 minutes	Open 7.5 minutes	15 min/day

Key Observations

- Both kits run the same valve duration per cycle at current stage even though Kit 1 has double the stands and double the water need, it also has double the emitters delivering simultaneously. The time equalises.
- As the crop grows the valve time increases by full fruiting stage both kits will need 22.5 minutes per cycle instead of 7.5. The operator must adjust valve timing as the crop matures.
- Kit 1 Stand 1 tank draw: 300L per cycle from a 2,000L tank lasts approximately 6–7 cycles before refill is needed.
- Kit 7 Stand 2 tank draw: 150L per cycle from a 2,000L tank lasts approximately 13 cycles before refill is needed.
- At fruiting stage Stand 1 will only last 4–5 cycles (approximately 2 days) before needing a refill daily monitoring becomes critical.

E6. Pumping Sessions Required Per Day

Kit 1 Stand 1 (2,000L Storex Capacity)

Growth Stage	Daily Need	Stand Lasts	Sessions Needed
Transplant	600L/day	~3 days	1 session every 3 days
Vegetative	900L/day	~2 days	1 session every 2 days
Fruiting	1,800L/day	~1 day	1 session DAILY

Kit 7 Stand 2 (2,000L Storex Capacity)

Growth Stage	Daily Need	Stand Lasts	Sessions Needed
Transplant	300L/day	~6 days	1 session every 6 days
Vegetative	450L/day	~4 days	1 session every 4 days
Fruiting	900L/day	~2 days	1 session every 2 days

Main Tower Buffer (10,000L)

Growth Stage	Combined Daily Need	Tower Lasts
Transplant	900L/day	~11 days
Vegetative	1,350L/day	~7 days
Fruiting	2,700L/day	~3–4 days

At current transplant stage, one full tower fill per week is sufficient to maintain the entire overhead system. As crop progresses to fruiting stage, plan for tower top-up every 3 days.

E7. Irrigation Schedule

Time	Activity	Notes
6:30 – 7:00 AM	First irrigation cycle	Cooler temperatures reduce evaporation loss
11:00 – 11:30 AM	Optional mid-morning cycle	Only if soil moisture is visibly low
4:30 – 5:00 PM	Second irrigation cycle	Avoid late evening to reduce fungal disease risk

Avoid irrigating between 11:30 AM and 3:30 PM during hot dry periods. On rainy days, skip or significantly reduce irrigation. Check drip lines after rain for silting or displacement.

Section F Block Moulding Water Usage

F1. Overview

Block moulding is an active construction operation at WGNI Farm supervised by Mr. Kitab (Farm Supervisor). The operation runs 6 days per week from 7:00 AM to 6:00 PM using a motorised petrol-engine block moulding machine. Water is drawn directly from the 6 x 5,000L base tanks via a hose pipe connected to the moulding site approximately 20–50 metres from the tanks. There is currently no dedicated water point at the block moulding site. The moulding team draws water as needed throughout the day.

F2. Block Moulding Output & Water Consumption

Item	Quantity / Value
Blocks produced per day	~375 blocks
Cement bags used per day (typical)	~14 bags
Water per bag of cement (standard ratio)	~20 litres
Estimated water for mixing per day	~280 litres
Estimated curing water per day	~50–75 litres
Total estimated water draws per day	~330–355 litres
Weekly water draw (6 operating days)	~2,000–2,130 litres
Monthly water draw	~8,000–8,520 litres

F3. Water Draw Procedure Block Moulding

1. Before drawing water, the operator or supervisor must visually check the base tank level if the combined base tank level appears critically low (at or below 5,000L combined), do not draw for block moulding. Notify the Farm Manager immediately.
2. Connect the hose pipe securely to the base tank outlet.
3. Open the outlet valve slowly do not open fully at once to avoid pressure surge.
4. Draw only the volume needed for the immediate mixing batch. Do not leave the hose running unattended.
5. Close the valve fully and disconnect the hose after each draw session.
6. Record the estimated volume drawn in the daily water usage log.

F4. Water Quality Note

Water drawn directly from the base tanks for block moulding does not pass through the filtration unit. This is acceptable for construction use. However, ensure the hose pipe connection point is clean and the tank outlet is covered when not in use to prevent contamination that could affect the greenhouse water supply downstream.

F5. Curing Water

After blocks are moulded, they require curing light watering over 2–3 days to strengthen. Curing water is also drawn from the base tanks via the same hose pipe. Estimate an additional 50–75 litres per day during active curing periods. This must also be recorded in the daily water usage log.

F6. Block Moulding Fuel Reference

Item	Reference
Machine type	Motorised petrol engine
Fuel used per 14-bag session (13 Apr 2026)	5 litres
Fuel cost per session (at N1,300/L)	~N6,500
Supervisor	Mr. Kitab (Farm Supervisor)
Operating hours	7:00 AM – 6:00 PM
Operating days	6 days per week

Section G Fuel Management

G1. Transfer Pump Fuel Estimates

Mode / Session	Volume	Est. Time	Petrol	Cost @ N1,300
Mode 1 Fill main tower	10,000L	~185–220 min	~1.90L	~N2,470
Mode 2 Fill Stand 1 only	2,000L	~35–40 min	~0.38L	~N494
Mode 3 Fill Stand 2 only	2,000L	~35–40 min	~0.38L	~N494
Fill Stand 1 + Stand 2 (no tower)	4,000L	~70–80 min	~0.75L	~N975
Full system fill	14,000L	~4 hrs	~2.6L	~N3,380
Recommended budget (with buffer)			3.0L	N3,900

G2. Monthly Fuel Budget Projection (Transfer Pump Only)

Pumping Frequency	Monthly Petrol	Monthly Cost
Weekly full tower fill (current transplant stage)	~8L	~N10,400
Every 3 days (vegetative stage)	~19L	~N24,700
Every 2 days (early fruiting)	~28L	~N36,400
Daily full fill (peak fruiting)	~57L	~N74,100

Note: Budget separately for generator fuel for intake pumping and petrol for the block moulding machine.

G3. Fuel Storage Rules

- Store petrol in a clearly labelled, sealed approved container.
- Keep fuel away from direct sunlight and at least 5 metres from any engine or open flame.
- Do not store more than 10 litres on-site at any time without proper storage provision.
- Log every fuel purchase and usage in the daily operations record.
- Always purchase fuel from a verified station adulterated fuel will damage the pump engine.
- Never mix fuel containers between transfer pump, generator, and block moulding machine.

Section H Daily Water Balance & Site Usage Log

H1. Daily Water Balance Check

Check	Target	Action if Below Target
Base tank level	Above 5,000L combined	Activate intake pumps immediately. Pause all non-greenhouse draws.
Solar pump / Eco flow status	Active and pumping	Check controller. Escalate if fault persists.
Generator status	Fuel sufficient, running stable	Refuel before starting operations.
Main tower level	Above 3,000L	Schedule pumping session for that day.
Stand 1 level	Above 500L	Fill Stand 1 before morning irrigation.
Stand 2 level	Above 500L	Fill Stand 2 before morning irrigation.
Crop soil moisture	Moist at drip emitter point	Adjust irrigation cycle duration accordingly.

H2. Daily Water Usage Log

The following must be recorded every day by the pump operator, construction supervisor, or farm supervisor:

Field	Entry
Date	
Weather condition	Sunny / Cloudy / Rainy
Solar pump / Ecoflow status (AM)	Active / Fault / Low output
Generator run hours	_____ hours
Base tank level (estimated AM)	_____ litres
Main tower level (AM)	Full / Three-quarter / Half / Low / Empty
Stand 1 level (AM)	Full / Half / Low / Empty
Stand 2 level (AM)	Full / Half / Low / Empty
Transfer pump mode used	Mode 1 / 2 / 3 / None
Transfer pump session start time	
Transfer pump session stop time	
Volume pumped to greenhouse system (est.)	_____ litres
Transfer pump petrol used	_____ litres
Filter status	Normal / Backwash performed
Irrigation cycles completed	Morning / Midday / Evening
Kit 1 valve open time cycle 1	_____ minutes
Kit 1 valve open time cycle 2	_____ minutes
Kit 7 valve open time cycle 1	_____ minutes
Kit 7 valve open time cycle 2	_____ minutes

Block moulding water draw (est.)	_____ litres
Block moulding cement bags used	_____ bags
Block moulding petrol used	_____ litres
Blocks produced today	_____ units
Other site water draw (poultry/fish/construction)	_____ litres
Total estimated water drawn from base tanks today	_____ litres
Estimated closing base tank level	_____ litres
Issues observed	
Supervisor name	

H3. Water Usage Reference 13 April 2026 Example

The following is a recorded example session from 13 April 2026 for reference and training purposes:

Item	Recorded	Calculated
Opening base tank level	30,000L (assumed full)	
Transfer pump fuel used	5 litres petrol	~5.5 hrs pumping / ~16,500L transferred
Block moulding fuel used	5 litres petrol	
Cement used block moulding	14 bags	
Block moulding water draw (est.)	~280L mixing + ~75L curing	~355L total
Total water drawn from base tanks		~16,855L
Estimated closing base tank level		~13,145L
Fuel price on this date	N1,257/L	Slightly below N1,300 working rate
Transfer pump fuel cost (5L)	N6,285	
Block moulding fuel cost (5L)	N6,285	

Section I Water Management Guidelines

11. Water Priority Hierarchy

When base tank levels are low, water allocation must follow this priority order:

Priority	Activity	Reason
1 Highest	Greenhouse drip irrigation	Crop survival and yield non-negotiable
2	Poultry and fishpond	Animal welfare requires consistent supply
3	Block moulding mixing	Construction can pause without permanent loss
4 Lowest	Block moulding curing	Can be deferred 24–48 hours if necessary

12. Water Conservation Rules

- Never run drip lines while it is actively raining.
- Reduce irrigation duration by 30–40% on overcast or cool days.
- Inspect drip tape weekly for blocked or burst emitters a burst emitter wastes water and causes uneven crop growth.
- Do not allow any overhead tank to overflow monitor fill time and stop the pump promptly.
- Never pump from base tanks below the 5,000L safety threshold.
- Close all en-route valves properly after each session to prevent unintended drainage.
- All hose pipe connections at the base tanks must be closed and capped when not in use.
- Block moulding team must not leave the hose pipe running unattended at any time.

13. Seasonal Water Management

Season	Period	Adjustments
Dry season	Nov – Mar	Increase irrigation frequency. Monitor base tanks daily. Supplement solar with electric surface pump as needed. Consider adding a third daily irrigation cycle at peak dry heat.
Rainy season	Apr – Oct	Reduce irrigation on rain days. Inspect drip lines weekly for silting and displacement. Increase filter backwash frequency. Block moulding curing may be reduced or paused on heavy rain days.

14. Water Quality Monitoring

- Observe filter outlet water clarity at the start of every pumping session.
- If water appears cloudy or discoloured, do not pump to the greenhouse perform backwash first, then reassess.
- Inspect pre-filter cartridges every two weeks. Replace when visibly clogged.
- If crop shows unexplained tip burn, leaf curl, or unusual marks, collect a water sample and report to the Farm Manager.
- Keep all tank inlets always covered when not actively filling.
- Block moulding team must ensure spent cement water and washdown water does not flow toward the base tanks or borehole head.

15. System Capacity Summary

Storage Level	Capacity	Notes
Borehole	Continuous	Primary source subject to pump availability
Base tanks (6 units)	30,000L	Primary buffer replenished daily by Ecoflow solar system
Main water tower	10,000L	Primary overhead reservoir weekly fill at current stage
Stand 1 (Storex)	2,000L	Feeds Kit 1, 2, 3, 6 fill every 2–3 days currently
Stand 2 (Storex)	2,000L	Feeds Kit 4, 5, 7 fill every 5–6 days currently
Total system capacity	44,000L	Base + overhead combined

Section J Safety Rules

- Never operate any petrol pump or the block moulding machine in an enclosed or poorly ventilated space exhaust fumes are hazardous.
- Do not refuel any engine while it is hot or running. Allow a minimum of 5 minutes cooling before refuelling.
- Only designated and trained farm personnel should operate the transfer pump, adjust filter manifold valves, or access the main tower platform.
- Do not climb the main water tower unless there is a specific maintenance need and a second person is present on the ground.
- Always wear footwear near the pump station, filter unit, base tank area, and block moulding site.
- Keep the water station canopy area clear of debris, tall grass, and standing water always.
- Do not allow unauthorised personnel near the borehole head, Ecoflow control panels, or filter unit.
- In the event of an electrical fault at the solar controller, generator panel, or booster pump panel, isolate power immediately and notify the Farm Manager do not attempt to repair without authorisation.
- Store all fuel containers transfer pump, generator, and block moulding machine in separate clearly labelled sealed containers. Never mix fuel containers.

Section K Escalation & Fault Reporting

Stop operations immediately and notify the Farm Manager if any of the following occur:

Fault	Immediate Action
No water flow after two priming attempts	Stop pump. Do not attempt a third start. Report immediately.
Ecoflow controller showing fault code	Do not reset without guidance from the Farm Manager.
Generator fails to start or trips under load	Check fuel and oil. If unresolved within 5 minutes, report.
Electric surface pump fails despite generator running	Check cable connection. If unresolved, report.
Filter booster pump fails to start or trips repeatedly	Check power supply. If unresolved within 5 minutes, report.
Visible leak at any pipe joint, vessel, valve, or tank	Stop pumping. Isolate the affected section. Report.
Any overhead tank overflows	Check and repair float valve. Report if absent or faulty.
Base tanks drop below 5,000L combined	Suspend all non-greenhouse water draw immediately. Prioritise refill.
Main tower shows visible structural movement or corrosion	Do not pump to tower. Report immediately.
Drip lines showing uneven flow or burst tape in active kit	Stop irrigation to affected kit. Report to Farm Manager.
Water discolouration persists after backwash	Do not pump to greenhouse. Report immediately.
Petrol engine making unusual knocking or rattling sounds	Stop engine immediately. Do not restart. Report.
Block moulding machine breakdown	Notify Mr. Kitab and Farm Manager. Log downtime and blocks lost.
Block moulding water draw causing base tank to approach 5,000L	Stop block moulding draw immediately. Prioritise greenhouse. Report.

Section L Review & Version Control

Version	Date	Prepared By	Changes
1.0	April 2026	Farm Operations Management	Initial issue full system documentation including greenhouse, block moulding, valve timing, and all site water usage

This SOP is subject to review whenever there is a significant change to the water infrastructure, crop mix, greenhouse kit configuration, pump equipment, valve timing, or construction activity. The Farm Manager must approve all revisions before implementation. A printed copy of the current version must be always kept at the pump station and at the block moulding site.